Member 3: Structure-Preserving Augmentation Report Handoff

NNDL Project – Member 3 Materials

Abstract

This handoff contains the related-work and method drafts, three current-status bullets, the ablation plan, and standalone figures. Logical placement is 2.4, 3.5, 4.3, and 5.3 in the group report. Full augmentation generation is complete; classification evaluation and LPIPS remain pending.

2 Related Work

2.4 Structure Preservation and Structure-Related Metrics

Style augmentation is useful only when the generated image retains the evidence supporting its content label. Gatys *et al.* separate content and style using convolutional feature representations and feature correlations [2]. AdaIN provides a feed-forward alternative by matching channel-wise feature means and variances [3]. Its original formulation also permits spatial control. Our project uses an edge-derived spatial strength map as a lightweight content-preservation heuristic; we do not claim to introduce spatial control itself or to obtain a semantic foreground mask.

Image gradients provide inexpensive local structural cues. Sobel derivatives approximate intensity changes along two axes [4]; their magnitude can indicate contours, but also responds to background texture. We therefore use a smoothed, normalized Sobel response to guide generation, while reserving Canny edges [1] for a separate consistency analysis. This distinction keeps the generation map and evaluation procedure explicit. Neither detector establishes whether an edge is important for the object class.

We examine generated images with complementary measures. LPIPS compares images through deep feature representations and was evaluated against human perceptual judgments [5]; it is not a dedicated contour-preservation score. Our project-defined EdgeScore is the F1 overlap of fixed-threshold Canny edge maps, measuring how many content edges survive and how many generated edges agree with them. It is not a metric proposed by the Canny paper. Exact edge overlap is sensitive to small displacements and threshold choices, whereas LPIPS also responds to intended appearance changes. We interpret both alongside qualitative examples and the final classification experiment. Lower perceptual distance or higher edge overlap alone cannot demonstrate better cross-domain recognition: an almost unchanged photo may obtain favorable similarity scores while adding little style diversity.

3 Method

3.5 Structure-Preserving Style Augmentation

We reuse member 2's frozen AdaIN encoder and decoder [3] and independently implement the structure map, spatial strength rule, and offline generation pipeline. The content image I_c is drawn only from Photo Training. Its style image I_s is read from the same frozen external pool and content–style pairing used by ordinary AdaIN.

Structure map. After resizing the shorter content side to 512 pixels, we form grayscale intensity $Y = 0.299R + 0.587G + 0.114B$ and smooth it with a Gaussian kernel of standard deviation 1, giving $J = G_1 * Y$. Let K_x and K_y be the 3×3 Sobel kernels, scaled by $1/8$. The gradient response is

$$g = \sqrt{(K_x * J)^2 + (K_y * J)^2}. \tag{1}$$

Responses at most 10^{-6} are treated as zero. We set s to the 95th percentile of g; if that value is at most 10^{-6}, we use the maximum response instead. With $\epsilon = 10^{-6}$, the soft structure map is

$$\hat{g} = \mathrm{clip}_{[0,1]}\left(\frac{g}{\max(s, \epsilon)}\right),$$
$$M = \mathrm{clip}_{[0,1]}(G_2 * \mathrm{MaxPool}_{3\times3}(\hat{g})). \tag{2}$$

Maximum pooling uses stride 1 and preserves image size. It broadens thin responses; the second Gaussian, with standard deviation 2, softens their transition. Constant images receive a zero map. Generation uses this continuous response, rather than a Canny mask or a learned saliency model.

Adaptive strength and feature mixing. For base style strength α_0 and protection parameter β, we define

$$\alpha(x, y) = \alpha_0[1 - \beta M(x, y)], \qquad \alpha_0, \beta \in [0, 1]. \tag{3}$$

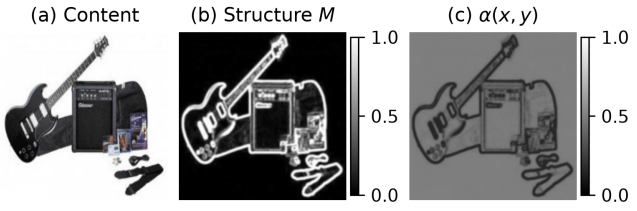


Figure 1: Actual Photo Training example: content image, soft structure response M, and spatial strength $\alpha(x, y)$ for $\alpha_0 = 0.5$, $\beta = 0.7$. Both map color scales are fixed to $[0, 1]$. Brighter M produces lower local style strength. The map also responds to background contours.

The map is area-averaged to the content encoder's `relu4_1` resolution, giving α_f. Let $F_c = E(I_c)$ and $F_s = E(I_s)$. The decoder input and output are

$$F_{\text{mix}} = \alpha_f \odot \text{AdaIN}(F_c, F_s) + (1 - \alpha_f) \odot F_c, \quad (4)$$

$$I_{\text{sp}} = D(F_{\text{mix}}). \quad (5)$$

The one-channel map broadcasts over the 512 feature channels. Strong structure responses reduce the style contribution. At $\beta = 0$, the rule reduces to ordinary constant-strength AdaIN; this equivalence is verified in the implementation. At $\alpha_0 = 0$, the result is an encoder–decoder reconstruction, not an exact copy of the input. Figure 1 visualizes the actual maps; the complete pipeline is shown in Fig. 2.

Offline delivery and limitations. The initial full set uses $\alpha_0 = 0.5$, $\beta = 0.7$ and contains one output for each of the 1,349 source training images. The latter parameter is an initial setting, not a selected optimum. Images are written to `data/aug_sp/`; a manifest records the output path, label, original, style, α_0, and β. Geometry, frozen weights, content–style pairing, and JPEG quality 95 are held fixed relative to the matched AdaIN reference. Output dimensions follow the same decoder behavior in both methods. The map is an edge-based proxy and cannot guarantee object preservation after decoding. Since the rule also lowers mean style strength, a uniform control with matched mean strength is needed to isolate the effect of spatial placement.

4 Progress / Current Status

4.3 Structure-Preserving Method Progress

- *Implementation completed.* We implemented the soft Sobel structure map, adaptive strength $\alpha_0(1 - \beta M)$, feature-level AdaIN mixing, batch generation, diagnostic panels, and parameter sweeps on top of the pinned member-2 interface. Twelve automated tests passed, including constant-image handling, parameter boundaries, recovery from corrupt outputs, manifest compatibility, and equivalence between $\beta = 0$ and ordinary AdaIN.

- *Full training augmentation delivered locally.* The generator produced 1,349 real Photo Training augmentations in `data/aug_sp/` at $\alpha_0 = 0.5$, $\beta = 0.7$. All outputs were decoded and checked for labels, frozen pairings, input/output hashes, and manifest consistency. Two-column and six-column manifests are available; the latter is directly readable by member 4. Seven class examples with structure and strength maps are included in the handoff.

- *Analysis prepared; classification evaluation pending.* The full set has EdgeScore 0.3436 ± 0.0628 over 1,349 images, where the spread is across images, not classifier seeds. A five-value β pilot was completed on seven dog images; it demonstrates the generation workflow and is not a full ablation. LPIPS and augmented-group classification results remain pending. Code, figures, manifests, audit records, and the ablation plan are ready for integration.

5 Next Steps and Plan for Completion

5.3 Ablation and Analysis Plan

Protection-strength ablation. We will hold $\alpha_0 = 0.5$ fixed and generate $\beta \in \{0, 0.3, 0.5, 0.7, 1\}$ variants for the same 1,349 Photo Training images. The style pool, content–style pairing, map settings, input sizes, augmentation count, and image encoding will remain unchanged. The $\beta = 0$ case is ordinary AdaIN and serves as the implementation and experimental control. Member 3 provides images and manifests; member 4 performs classifier training and evaluation under the shared protocol. The existing seven-image, dog-only pilot checks execution and does not select an optimal β.

Structure and appearance analysis. For every variant, we will compute the fixed-threshold Canny EdgeScore and LPIPS, using the same preprocessing as Section 3.6. We will report means, image-level standard deviations, valid counts, and per-image differences relative to $\beta = 0$. Empty-edge cases will remain undefined and their number will be disclosed. LPIPS is planned and has not yet been computed. Qualitative panels will use fixed $[0, 1]$ map scales and matched content/style pairs across classes. We will inspect thin structures, textured backgrounds, and failures where the map protects irrelevant edges.

Reduced-strength confound. Increasing β decreases average style strength as well as adding spatial modulation. A recommended additional control is uniform AdaIN

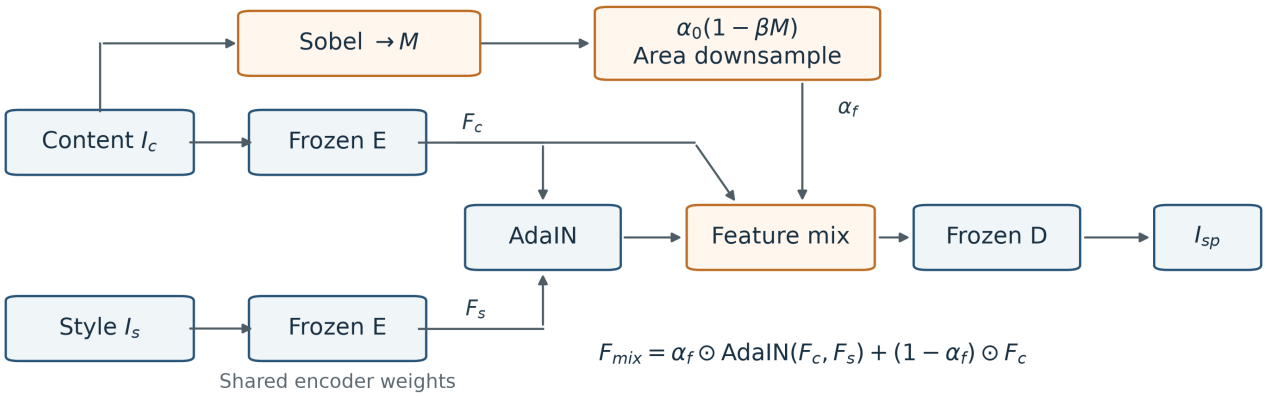


Figure 2: Structure-guided feature mixing. The structure branch uses the content image only; the style reference changes feature statistics. Encoder and decoder weights remain frozen.

with per-image strength

$$\bar{\alpha}_i = \frac{1}{H_f W_f} \sum_{u,v} \alpha_{f,i}(u, v). \tag{6}$$

It uses the same content/style pair and decoder. This control is a proposed extension, not an output already produced by the current batch CLI. Comparing it with the spatial map would help separate the effect of where protection is applied from the effect of weaker stylization alone.

Classifier analysis and parameter discipline. Member 4 will use the shared ResNet-18, batch size 64, 660 optimizer steps, and final-checkpoint policy. A one-seed sweep can provide an exploratory budget; the primary comparison and any configurations used for a replicated claim will use seeds $\{0, 1, 2\}$. We will report Photo accuracy separately from the unweighted average of Art, Cartoon, and Sketch accuracies, with per-domain and relevant per-class results. Target-domain test data will not choose α, β, map settings, or checkpoints. Any selection rule will be fixed in advance or use source validation only; the initial $\beta = 0.7$ is retained as a predeclared setting. Higher EdgeScore or lower LPIPS will be treated as image analysis rather than proof of a classifier gain. A later alpha sweep may use $\{0.25, 0.5, 0.75, 1\}$ with its beta held fixed to avoid confounding both factors in the main beta comparison.

References

[1] John Canny. A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(6):679–698, 1986.

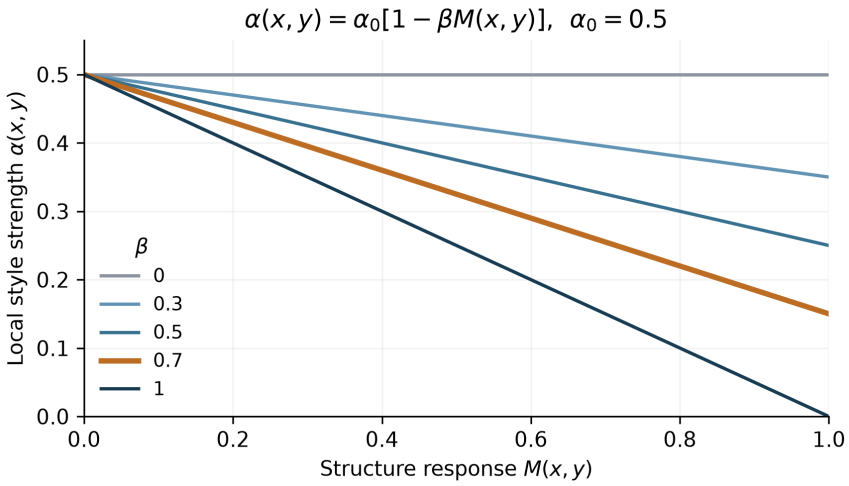
[2] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. Image style transfer using convolutional neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2414–2423, 2016.

[3] Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1501–1510, 2017.

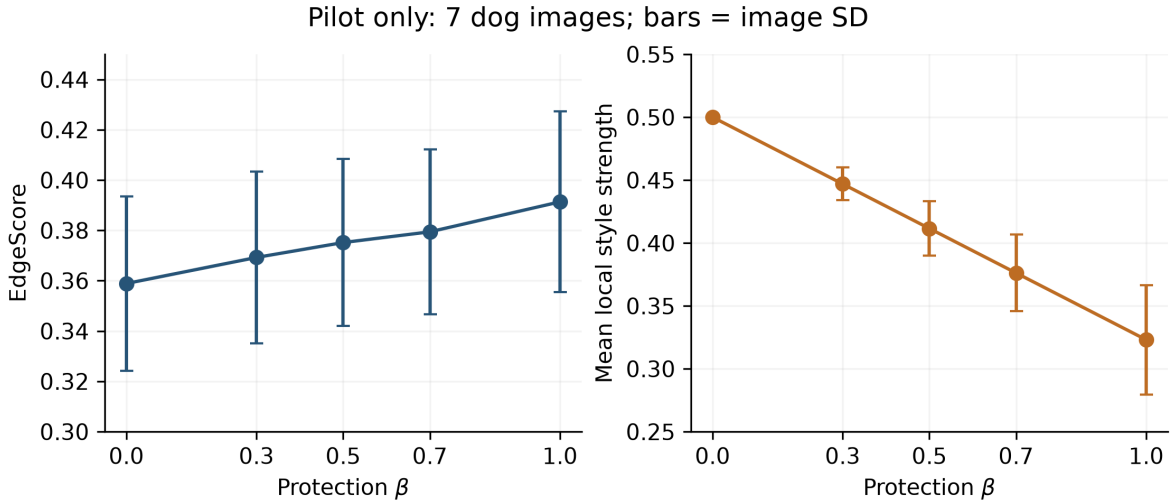
[4] OpenCV. Sobel derivatives. OpenCV Documentation. Accessed November 7, 2026.

[5] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, 2018.

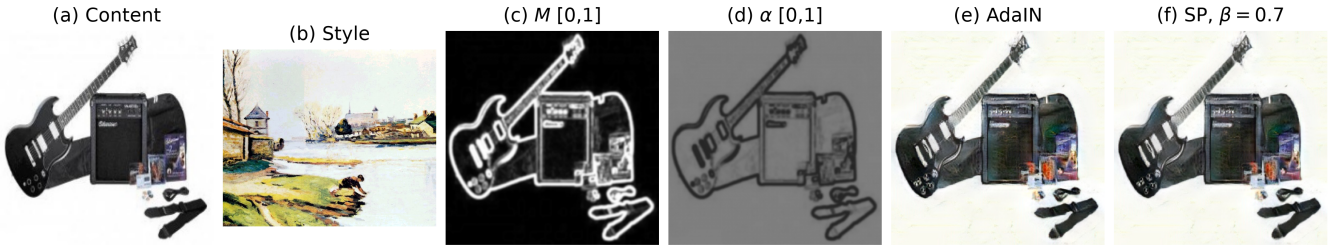
Supplementary Figure Material



Analytical strength rule. Fixed base strength is 0.5. Larger β lowers local strength where the structure response is stronger.



Pilot only. Seven dog images; error bars are image-level standard deviations. This pilot does not establish an optimal parameter or a classification gain.



Qualitative example. Content, style, structure response, spatial alpha, matched ordinary AdaIN, and structure-preserving output. This is not a classifier result.